

NLP UNIT-IV

- **Subject:**Artificial Intelligence/NLP
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Semantics

Definition

- Semantics is the branch of NLP that studies the meaning of words, phrases, and sentences independent of context.

Explanation

- Focuses on what a sentence means
- Concerned with literal meaning
- Answers: *“What does this sentence mean?”*

Example

The boy eats an apple.

- Meaning:
- *boy* → person
- *eats* → action
- *apple* → object

Applications

- Machine translation
- Question answering
- Information extraction

Pragmatics

Definition

- Pragmatics studies meaning in context, i.e., what the speaker intends to convey.

Explanation

- Depends on situation, speaker, listener
- Meaning may differ from literal sentence
- Answers: *“What does the speaker mean here?”*

Example

- Can you open the window?
- Literal meaning → Asking ability
- Pragmatic meaning → **Request to open the window**

Sentence + Context



Intended Meaning (Request)

Applications:

- Dialogue systems
- Conversational AI

Difference Between Semantics and Pragmatics

Semantics	Pragmatics
Literal meaning	Contextual meaning
Sentence-level	Discourse-level
Independent of context	Depends on context

Requirements for Semantic Representation

- Semantic representation is a formal way of representing meaning so that a computer can understand, reason, and infer from natural language.
- For an effective semantic representation, it must satisfy the following requirements:

1. Unambiguous
2. Express Relationships
3. Machine-Readable
4. Support Inference

1. Unambiguous

A semantic representation must have only one clear interpretation for a given meaning, removing ambiguity present in natural language.

Why Required?

- Prevents misunderstanding
- Essential for NLP tasks like translation and QA

Example (Ambiguous Sentence)

- I saw the man with a telescope.
- Two meanings:
- I used a telescope to see the man
- The man had a telescope

2. Express Relationships

Definition

- Semantic representation must clearly express relationships between entities, actions, and properties.

Explanation

- Meaning depends on who did what to whom
- Relationships include:
 - Agent
 - Action
 - Object
 - Time
 - Location

Example Sentence

- Ram gave a book to Sita.

3. Machine-Readable

Definition

- Semantic representation must be in a formal structure that computers can process, not free text.

Explanation

- Computers cannot understand natural language directly
- Meaning must be converted into:
 - Logic
 - Graphs
 - Structured symbols

Example Sentence

- The cat sleeps.

Machine-Readable Form

- sleep(cat)

4. Support Inference

Definition

- Semantic representation must allow deriving new facts from existing knowledge using reasoning.

Explanation

- Systems should not only store meaning
- They should infer implicit information
- Uses logical rules

Example

- $\text{Human}(\text{Ram})$
- $\forall x \text{ Human}(x) \rightarrow \text{Mortal}(x)$

Inference: $\text{Mortal}(\text{Ram})$

summary

Requirement	Purpose	Example
Unambiguous	Removes confusion	see(I, man, telescope)
Express relationships	Shows roles	give(Ram,Book,Sita)
Machine-readable	Computable form	sleep(cat)
Support inference	Derive new facts	Human → Mortal

First Order Logic (FOL)

- FOL definition

It is a formal system of representation used in NLP to represent the meaning of sentences using predicates, variables, constants, and quantifiers, enabling machines to reason and infer new knowledge.

Natural language is:

- Ambiguous
- Informal
- Not machine-understandable
- FOL converts natural language meaning into a **precise, unambiguous, and machine-readable form.**

Basic Components of First Order Logic

1.Constants

- Represent specific entities
- Examples: Ram, Sita, Apple
- Ram, Sita, Apple

2 .Variables

- Represent general entities
- Examples: x, y, z
- x, y

3 .Predicates

- Represent properties or relations
- Written as: predicate(argument1, argument2, ...)
- Examples:
- Student(Ram) Loves(Ram, Sita)

4. Quantifiers

A. Universal Quantifier ($\forall$)

- Means “for all”
- $\forall x \text{ Student}(x) \rightarrow \text{Human}(x)$
-  All students are humans

B. Existential Quantifier ($\exists$)

- Means “there exists”
- $\exists x \text{ Student}(x)$
-  There exists at least one student

examples

1. Simple Sentence Representation Using FOL

Example 1

Ram sleeps.

- FOL Representation:
- `sleep(Ram)`

2. Representing Relations in FOL

Example 2

Ram loves Sita.

- FOL Representation:
- `loves(Ram, Sita)`

3.Representing Properties in FOL

- Ram is a student.
- FOL Representation:
- $\text{Student}(\text{Ram})$

4.Representing Complex Sentences

- Every student studies NLP.
- FOL Representation:
- $\forall x \text{ Student}(x) \rightarrow \text{Studies}(x, \text{NLP})$
- Explanation
- x = any student
- If x is a student, then x studies NLP

5.Representing Existence

Some student passed the exam.

- FOL Representation:
- $\exists x \text{ Student}(x) \wedge \text{Passed}(x, \text{Exam})$

Combining Logic (AND, OR, NOT, IMPLIES)

Symbol	Meaning
$\wedge$	AND
$\vee$	OR
$\neg$	NOT
$\rightarrow$	IMPLIES

- Ram is a student and intelligent.
- $\text{Student}(\text{Ram}) \wedge \text{Intelligent}(\text{Ram})$

Description Logic (DL)

- Description Logic (DL) is a formal knowledge representation language used to describe concepts (classes), roles (relationships), and individuals (objects) in a structured and logical way.
- FOL is very powerful but complex
- DL provides:
 - Structured meaning
 - Easier reasoning
 - Efficient inference
- Widely used in ontologies and Semantic Web

Basic Components of Description Logic

1. Concepts (Classes)

- Represent categories or sets
- Written using capital letters
- Examples:
- Student Teacher Human

2. Individuals (Instances)

- Represent specific objects
- Written as constants
- Examples:
- Ram Sita

3 .Roles (Relations)

- Represent relationships between individuals
- Examples:
- teaches(Teacher, Student) parentOf(Ram, Sita)

Basic DL Representation:

Sentence:

- Ram is a student.

Description Logic Representation:

- Student(Ram)

Concept Hierarchy (Subclass Relationship):

Sentence:

- Every student is a human.

DL Representation:

- Student $\subseteq$ Human
-  *This represents class inclusion.*

3.Role (Relationship) Representation

Sentence:

- Ram studies NLP.

DL Representation:

- studies(Ram, NLP)

4.Combining Concepts and Roles

Sentence:

- A student is a human who studies a course.

DL Representation:

- $\text{Student} \subseteq \text{Human} \sqcap \exists \text{studies.Course}$

 *Explanation (as per textbook idea):*

- Student is a Human
- Student must study some course

Existential Quantification in DL

Meaning

- $\exists$ means “there exists”

Sentence:

- Ram studies some subject.

DL Representation:

- $\exists \text{studies.Subject}(\text{Ram})$

Explanation

- Ram has at least one studies relationship
- Subject is not specified

Example (Inference)

Given Knowledge:

- $\text{Student} \subseteq \text{Human}$ Student(Ram)

Inferred Knowledge: Human(Ram)

Difference between FOL&DL

Aspect	First Order Logic	Description Logic
Expressiveness	Very high	Moderate
Complexity	High	Lower
Reasoning	Difficult	Efficient
NLP Usage	Limited	Common

Syntax-Driven Semantic Analysis

Definition

Syntax-Driven Semantic Analysis is the process of **deriving the meaning of a sentence from its syntactic structure, where semantic interpretation is built compositionally from the meanings of words and phrases.**

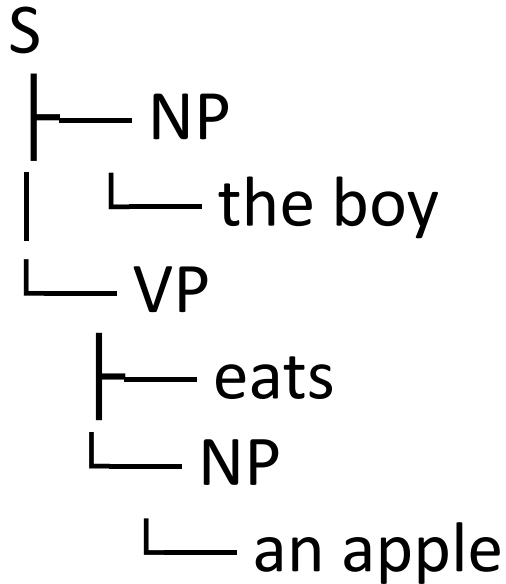
Syntax guides semantics

- Parse the sentence syntactically
- Use grammar rules
- Attach semantic meaning to each rule
- Combine meanings bottom-up

Basic Step-by-Step Example

Sentence

- The boy eats an apple.
- Step 1: Syntactic Parse



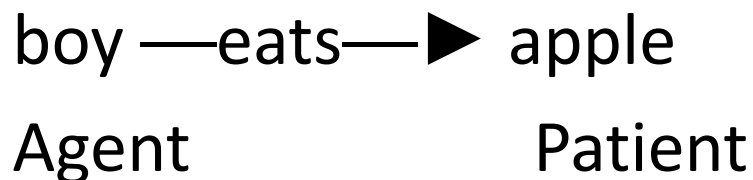
Step 2: Assign Semantic Roles

- Agent → the boy
- Action → eats
- Patient/Object → apple

Step 3: Semantic Representation

- eat(boy, apple)

diagram



Compositional Semantics

Definition

- Compositional Semantics states that:
- *The meaning of a sentence is derived from the meanings of its parts and the way they are combined.*

Example

- Ram loves Sita.
- Meaning of words: Ram, loves, Sita
- Structure: Subject–Verb–Object

Final Meaning

- love(Ram, Sita)

Syntax-Driven Semantics with Quantifiers

Sentence

- Every student studies NLP.

Syntactic Insight

- Subject: *every student*
- Verb: *studies*
- Object: *NLP*

Semantic Representation (FOL)

- $\forall x \text{ Student}(x) \rightarrow \text{Studies}(x, \text{NLP})$

Example Showing Role of Syntax Clearly

Sentence 1

- The dog chased the cat.

Meaning

- chase(dog, cat)

Sentence 2

- The cat chased the dog.

Meaning

- chase(cat, dog)
-  Same words
-  Subject position decides agent

Example with Quantifier

Sentence

- Every student studies NLP.

Syntax

- every student $\rightarrow$ universal group
- studies $\rightarrow$ verb
- NLP $\rightarrow$ object

Meaning (Logic form)

- $\forall x \text{ Student}(x) \rightarrow \text{Studies}(x, \text{NLP})$

Explanation in Simple Words

- For all students
- If someone is a student
- They study NLP

Example with Preposition

Sentence

- Ram cut the apple with a knife.

Syntax Roles

- Ram → Agent
- apple → Object
- knife → Instrument

Meaning

- cut(Ram, apple, knife)

Ambiguity Explained Simply Using Syntax

Sentence

- I saw the man with a telescope.

Meaning 1 (I used telescope)

- `see(I, man, using(telescope))`

Meaning 2 (Man has telescope)

- `see(I, man(with(telescope)))`

Why ambiguity happens

- Syntax allows two structures
- Each structure gives different meaning
- “Here ambiguity is identified by syntactic structure”

What are Semantic Attachments?

Definition

Semantic Attachments are semantic rules attached to syntactic grammar rules that specify how meaning is constructed during parsing.

Why Semantic Attachments are Needed?

Problem

- Syntax alone gives structure
- Semantics alone gives word meanings
- But sentence meaning needs both together

Solution

-  Attach meaning rules to grammar rules

Core Idea

- Whenever a grammar rule is applied, a semantic rule is also applied
- This means:
- Parsing + Meaning happen together
- Meaning is built bottom-up

Relationship with Syntax-Driven Semantics

Syntax	Semantic Attachment
Builds structure	Builds meaning
Uses grammar rules	Uses semantic rules
NP, VP, S	Logical meaning

- This is called compositional semantics.

Grammar Rule with Semantic Attachment

Grammar Rule

- $VP \rightarrow V NP$

Semantic Attachment

- $VP.sem = V.sem(NP.sem)$

Meaning

- VP meaning is created by:
 - Applying verb meaning to noun phrase meaning

Step-by-Step Construction

- Step 1: Parse sentence syntactically
- Step 2: Assign meaning to words
- Step 3: Apply semantic rules
- Step 4: Build full sentence meaning

Example

Sentence

- Ram eats rice.

Lexical Meanings

- Ram.sem = Ram eats.sem = eat rice.sem = rice

Apply Semantic Attachment

- VP.sem = eat(rice) S.sem = eat(Ram, rice)

Final Semantic Representation

- eat(Ram, rice)
- Diagram
- Ram —eats—► rice

Sentence: Ram eats rice

Step 1: Lexical Meanings (Word Meanings)

Each word in the sentence is first assigned a basic meaning.

Ram.sem = Ram eats.sem = eat rice.sem = rice

Explanation:

Ram is a person → represented as Ram

eats is an action → represented as eat

rice is an object → represented as rice

👉 At this stage, we only know individual word meanings, not sentence meaning.

Step 2: Syntactic Structure (Implicit)

The sentence follows the grammar pattern:

$S \rightarrow NP VP$

$VP \rightarrow V NP$

NP (subject) → Ram

V (verb) → eats

NP (object) → rice

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- Syntax tells us:
Ram is the doer
rice is the thing affected
This is the complete, unambiguous, machine-readable meaning of the sentence.

Step 3: Apply Semantic Attachment

Semantic rule attached to grammar rule:

$VP \rightarrow V \ NP \quad VP.sem = V.sem(NP.sem)$

Apply it here:

$VP.sem = eat(rice)$

Explanation:

The verb meaning eat

Is applied to the object rice

This forms the verb phrase meaning

👉 VP now means: *“eat rice”*

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Step 4: Build Sentence Meaning

Next grammar rule:

$S \rightarrow NP VP$ $S.sem = VP.sem(NP.sem)$

Apply it:

$S.sem = eat(Ram, rice)$

Explanation:

Subject Ram is added as the agent

Final meaning becomes:

Ram performs the action eat on rice

Step 5: Final Semantic Representation

$eat(Ram, rice)$

What this means:

$eat \rightarrow$ action

$Ram \rightarrow$ who performs the action (Agent)

$rice \rightarrow$ what is eaten (Object)

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What is Word Sense?

Definition

- A word sense is a specific meaning of a word in a particular context.
-  *Jurafsky & Martin define word sense as one of the possible meanings a word can have.*

Example

- Word: bank
- Bank (financial institution)
- Bank (side of a river)
-  Same word, different senses

Why Word Sense is Important

Explanation

- Many English words are ambiguous
- Correct meaning depends on context
- NLP systems must choose the right sense

Example

- He deposited money in the bank.
→ bank = financial institution
- He sat on the bank of the river.
→ bank = river side

Sense Relations (Introduction)

Definition

- Sense relations describe how meanings of words are related to each other.

Synonymy

- **Definition**

Two words are **synonyms** if they have **similar meanings**.

- **Examples**

big – large

car – automobile

Antonymy

- **Definition**
- Two words are **antonyms** if they have **opposite meanings**.
- **Examples**
- hot – cold
- happy – sad

Hypernym & Hyponym

Hypernym

- General term

Hyponym

- Specific term
- **Example**
- Animal → hypernym
- Dog → hyponym

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- **Meronymy/Holonymy (Part-Whole):** Connections based on composition.
 - **Meronym:** A part of something
 - (e.g., wheel is a meronym of car).
 - **Holonym:** The whole that contains parts
 - (e.g., car is a holonym of wheel).
- **Polysemy (Linked Senses):** A single word with multiple related meanings
- (e.g., mouth of a person vs. mouth of a river).

- **Homonymy (Unrelated Senses):** Words that share the same spelling or sound but have totally unrelated meanings
 - (e.g., *bat* the animal vs. *bat* the sports equipment).
- **Troponymy (Manner):** Verb relations indicating manner
 - (e.g., *whisper* is a way of *talking*).
- **Entailment (Implication):** Meaning of one verb implies another
 - (e.g., *snoring* entails *sleeping*).

Thematic Roles

Definition:

- Semantic roles, also known as thematic roles, describe the relationship between a verb and its arguments within a sentence. These roles provide a deeper understanding of the sentence by indicating how each entity (noun) is involved in the action described by the verb.

Why Thematic Roles are Needed?

- Syntax tells **position** (subject/object)
- Semantics tells **function**
- Helps answer:
 - Who did the action?
 - What was affected?
 - How was it done?

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Key concepts

- **Agent:** The entity that performs the action.
 - Example: *John* (agent) kicked the ball.
- **Patient:** The entity that is affected by the action.
 - Example: John kicked the *ball* (patient).
- **Instrument:** The entity used to perform the action.
 - Example: She cut the bread with a *knife* (instrument).
- **Experiencer:** The entity that experiences or perceives something.
 - Example: *Mary* (experiencer) heard a strange noise.

- **Theme:** The entity that is moved or the topic of the action.
 - Example: She gave the *book* (theme) to him.
- **Location:** The place where the action occurs.
 - Example: He stayed in the *house* (location).
- **Source:** The starting point of the action.
 - Example: She came from the *village* (source).
- **Goal:** The endpoint of the action.
 - Example: He walked to the *park* (goal).

Role	Meaning
Agent	Doer of action
Patient	Entity affected
Theme	Entity moved/changed
Experiencer	Feels or perceives
Instrument	Tool used
Location	Place
Source	Starting point
Goal	Destination

Role of Semantic Roles in Understanding Sentence Structure

- Semantic roles play a critical role in understanding sentence structure by:

Clarifying Relationships: They help in identifying how different entities in a sentence are related to the main action, providing a clear picture of who is doing what to whom, with what, and where.

Improving Disambiguation: By assigning roles, it becomes easier to disambiguate sentences that might be syntactically similar but semantically different.

Enhancing NLP Tasks: Many NLP tasks, such as information extraction, machine translation, and question answering, rely on understanding these roles to produce accurate and meaningful results.

What are Selectional Restrictions?

Definition

- Selectional restrictions are constraints imposed by a verb on the type of arguments it can take.
-  *Jurafsky & Martin explain that verbs restrict the semantic class of their arguments.*

Why Selectional Restrictions are Needed?

- Not all word combinations are meaningful
- Helps detect:
 - Semantic errors
 - Meaning violations

Simple Example

- Correct Sentence
- Ram eats rice.
- ✓ eat → requires food
- Incorrect Sentence
- Ram eats stone.
- ✗ stone is not food

Relationship Between Thematic Roles & Selectional Restrictions

Look at the below table ↓

- **Animate**
 - Refers to living beings that can act or feel
 - Examples: person, dog, child
 - Used for roles like Agent
- **Inanimate**
 - Refers to non-living objects
 - Examples: table, stone, book
 - Often used as Patient or Instrument
- **Edible**
 - Refers to things that can be eaten
 - Examples: rice, apple, food
 - Required by verbs like eat

Aspect	Thematic Roles	Selectional Restrictions	Explanation
Focus	Role in the event	Type constraint	Thematic roles tell what role an entity plays (doer, affected, tool), while selectional restrictions tell what kind of entity is allowed to play that role.
Example	Agent, Patient, Instrument	Animate, Inanimate, Edible	Animate = living beings (person, animal). Inanimate = non-living objects (stone, table). Edible = things that can be eaten (rice, apple).
Purpose	Meaning structure	Meaning validity	Thematic roles build the structure of meaning , while selectional restrictions ensure the sentence is semantically valid and meaningful .

What is Word Sense Disambiguation (WSD)?

Definition

- Word Sense Disambiguation (WSD) is the process of identifying the correct meaning (sense) of

Why WSD is important?

WSD is important in many natural language processing (NLP) applications because if word meanings are misidentified, the results can be problematic:

- Machine translation systems may produce incorrect translations due to misinterpretation of word meanings.
- Information retrieval systems might return irrelevant results because of semantic uncertainty or confusion.
- Question-answering systems require precise word meanings to provide accurate and relevant answers.
- a word based on its context.

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Primary Approaches to Word Sense Disambiguation

- 1. Knowledge-Based Methods
- 2. Supervised Learning Methods
- 3. Unsupervised Learning Methods

Broader WSD Challenges

1. Data sparsity
2. Sense granularity
3. Domain adaptation

Knowledge-Based WSD (Dictionary Based WSD)

- Uses dictionary or lexical resources
- Matches context words with word definitions

Example

- He sat on the bank near water.
- Context word water → river bank

Supervised WSD

- Uses labeled training data
- Learns which sense appears in which context

Example

- bank + money → financial sense
- bank + river → river sense

Unsupervised WSD

- No labeled data
- Groups contexts automatically
- Uses similarity patterns
- Simple Idea
- Similar contexts → similar meanings

What is Dictionary-Based WSD?

Definition

- Dictionary-based Word Sense Disambiguation is a method that identifies the correct meaning of a word by comparing dictionary meanings with the context words in the sentence or corpus.
-  It uses:
- Dictionary / glossary meanings
- Context words from sentence or corpus

Steps in Dictionary-Based WSD

1. Identify the ambiguous word
2. Get all meanings from dictionary
3. Collect context words from sentence/corpus
4. Compare dictionary meanings with context
5. Select the best matching sense

Example 1

Sentence

- *The fisherman sat on the bank near the river.*

Context Words

- fisherman
- river
- sat

Matching with Dictionary

- River bank meaning contains: *river, shore*

✓ **Selected Sense:**
Bank = River Side

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Example2

Dictionary Meanings Of Plant

- Plant¹ → living organism (tree, flower)
- Plant² → factory (industry, machinery)

Corpus Sentence (Plant – Meaning 1)

Sentence

- *The plant needs water and sunlight.*

Context Words

- water
- sunlight

✓ **Selected Sense:**
Plant = living organism

Corpus Sentence (Plant – Meaning 2)

Sentence

- *The company built a new plant in Hyderabad.*

Context Words

- company
- built

✓ Selected Sense:

Plant = factory

Flow Chart

Sentence / Corpus

Context Words

Dictionary Meanings

Best Match

Correct Sense



Common Tools and Resources

- **WordNet:** The primary lexical database for English WSD.
- **NLTK (Natural Language Toolkit):** A Python library that provides a built-in implementation of the Lesk algorithm.

Limitations

- Depends on dictionary quality
- Context overlap may be weak
- Not very accurate for complex text
- Fails for metaphors

Example:

- *The news hit him hard.*
(dictionary meaning fails)

Supervised Word Sense Disambiguation (WSD)

- Supervised Word Sense Disambiguation (WSD) is an approach in Natural Language Processing (NLP) that uses machine learning to determine which meaning (sense) of a word is intended in a specific context.
- It relies on a sense-tagged corpus—a large collection of text where each instance of an ambiguous word has been manually labeled with its correct meaning.

Key Components of Supervised WSD

- **Sense Inventory:** A predefined list of meanings for words, often sourced from lexical databases like [WordNet](#).
- **Training Corpus:** A dataset where every ambiguous word is tagged with a sense from the inventory (e.g., SemCor).
- **Features:** Information extracted from the context to help the classifier:
 - **Collocations:** Specific words immediately to the left or right of the target word.
 - **Bag-of-Words:** All words appearing within a fixed "window" around the target.
 - **Part-of-Speech (POS):** Grammatical tags (noun, verb, etc.) of neighboring words.

General Workflow

- **Preprocessing:** Extract features from the sense-tagged training data.
- **Training:** A machine learning algorithm (e.g., Naive Bayes, SVM, or Decision Trees) learns a model that maps these features to specific word senses.
- **Disambiguation (Classification):** For a new, unseen sentence, the model calculates the probability of each possible sense and assigns the one with the highest score.

Comprehensive Example: The word "Bank"

- Suppose we have two senses for "bank":
- Sense 1 (Financial): A place where money is kept.
- Sense 2 (River): The land alongside a body of water

Input Sentence: "I went to the **bank** to deposit my check." The target word is "bank".

Context Extraction: Identify surrounding words.

Features: "deposit," "check," "went".

Feature Matching : Compare with training data. Training data shows "deposit" frequently appears with **Sense 1**.

Classification : Apply algorithm (e.g., Naive Bayes).The model calculates: $P(\text{Sense 1})$

Result : Final Output **Sense 1 (Financial Institution)**.

Popular Supervised Algorithms

Naive Bayes : Predicts the most likely sense based on the conditional probability of context words.

Support Vector Machines (SVM) : High-performance classifiers that find the optimal boundary between different senses.

Decision Lists : A sequence of weighted "if-then-else" rules (e.g., "if 'river' is nearby, then Sense 2").

INPUT RAW DATA

Labeled Data



Labels



Supervised Machine Learning

Model is trained using labeled data and pre-processed to predict outcomes and classify data accurately.

Processing

Algorithm

OUTPUT



Elephant



Camel



Camel

- **Pros:** Generally achieves higher accuracy than unsupervised or knowledge-based methods because it learns directly from human-annotated examples.
- **Cons:** Suffers from the "knowledge acquisition bottleneck"—it is expensive and time-consuming to create the large, manually tagged datasets required for training.

Thesaurus-Based Word Sense Disambiguation (WSD)

- Thesaurus-based Word Sense Disambiguation is a knowledge-based method used to find the correct meaning of an ambiguous word by using a thesaurus (like *Roget's Thesaurus*).
-  Instead of using dictionary definitions, it uses semantic categories (topics or themes) such as:
 - Finance
 - Education
 - Emotion
 - Nature

How is it Different from Dictionary-Based WSD?

- **Dictionary-based WSD** → compares word meanings (definitions)
- **Thesaurus-based WSD** → compares categories or topics
-  This method assumes:
- Words used together in a sentence usually belong to the same topic or category.

Main Method: Walker's Algorithm

Step 1: Find Possible Senses

- Take the ambiguous word
- List all its possible meanings
- Find the category of each meaning from the thesaurus

Step 2: Look at Context Words

- Observe words near the ambiguous word in the sentence

Step 3: Match Categories

- Check which sense's category matches most context words
- Each match = 1 point

Step 4: Choose the Best Sense

- The sense with highest score is selected

- Example: Word “Interest”

Sentence

The bank offered a high interest rate on my savings.

Sense	Category	Context Words Matching Category	Score
Financial Interest	Finance / Money	bank, rate, savings	3
Curiosity / Attention	Mental / Emotion	none	0

- Final Result
- ✓ Financial sense is selected
👉 Because more context words belong to the Finance category

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Bootstrapping for Word Similarity

- In Natural Language Processing (NLP), bootstrapping for word similarity is a semi-supervised learning method that starts with a small set of known words and gradually learns more similar words automatically.
-  It learns step by step, using very little initial information.

Main Idea

- Start with a few correct examples, learn from them, find more similar words, and repeat the process.

Core Mechanism of Bootstrapping (Easy Cycle)

Start Small → Learn → Expand → Repeat

Step 1: Seed Selection

- Begin with a small set of reliable word examples, called seed words or seed pairs.
- These seeds already have a known relationship.

Example

- car – automobile
- teacher – instructor
- These pairs clearly have similar meanings.

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Step 2: Pattern Extraction

- Search a large, unlabelled text corpus for sentences where these seed words appear.
- Identify common language patterns around them.

Example Pattern

- "X is a type of Y"
- Sentence
- A car is a type of vehicle.

Step 3: Expansion

- Use the discovered patterns to find new words that fit the same pattern.
- Example
- A bike is a type of vehicle.
-  Now bike is added as a new similar word.

Step 4: Labeling and Re-training

- Select only the most confident new words.
- Add them to the seed list.
- Repeat the process with the expanded set.
- 👉 The system becomes more accurate with each round.

Flow diagram

Seed word: car



Find patterns in corpus



Discover: bike, truck



Add to seed list



Repeat learning

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1. Dictionary-Based Bootstrapping

What it Means

- Dictionary-based bootstrapping uses dictionary definitions to learn word meanings and expand semantic categories.

How It Works (Step-by-Step)

Step 1: Initial Seeds

- Start with a few words that clearly belong to a category.

Example:

- Category: *Fruit*
- Seed words: *apple, mango*

Step 2: Iterative Labeling

- Look into dictionary definitions (glosses) of these seed words.
- If new words appear frequently in these definitions, they are added to the same category.

Example:

- Definition of apple → “a sweet fruit”
- New word found: *fruit* → added

Step 3: Sense Embedding (Simple Meaning)

- Modern systems treat dictionaries as reliable text sources.
- They learn different meanings (senses) of a word and represent them separately.
- This creates sense embeddings, which are better than simple word vectors.

Example:

- *bank* (money sense)
- *bank* (river sense)

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Step 4: Pattern Extraction

- The system finds common definition patterns like:
- "X is a type of Y"

Example:

- "Apple is a type of fruit"
- These patterns are reused to find new correct words.

2. Thesaurus-Based Bootstrapping

What it Means

- Thesaurus-based bootstrapping uses word hierarchies and relations from a thesaurus (like WordNet).

Key Idea

- Words are similar if they are close to each other in a hierarchy.

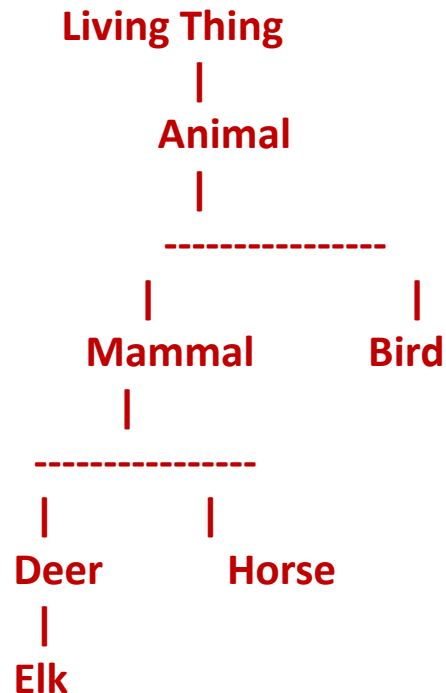
Least Common Subsumer (LCS)

- LCS is the closest common parent of two words in the hierarchy.
-  Closer LCS = higher similarity

(i) Path-Based Similarity (Very Simple)

Concept

- Words are arranged in a hierarchical tree
- Shorter path distance = higher similarity
- Similarity depends on how close two words are in the tree



How Path-Based Similarity Works (Using the Tree)

Example 1: Deer and Elk

- Path:
Deer → Elk (very short)

Least Common Subsumer (LCS): Deer

-  High similarity

Example 2: Deer and Horse

- Path:
Deer → Mammal → Horse

LCS: Mammal

 Moderate similarity

Example 3: Deer and Bird

- Path:
Deer → Mammal → Animal → Bird

LCS: Animal

 Low similarity